MILITARY SPECIFICATION

SURVIVAL KIT CONTAINER, AIRCRAFT SEAT WITH OXYGEN - RSSK-1A

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This detail specification together with the general specification cover requirements for the procurement of a RSSK-1A rigid seat survival kit.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-S-81018	Survival Kit Container, Aircraft Seat with Oxygen; General Specification for

DRAWINGS

Naval Air Systems Command

67A73D1	Parts Index, Survival Kit System - RSSK - 1A

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(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government quality control representative. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code 1051) 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification - The RSSK-1A rigid seat survival kit furnished under this specification shall be a product which has been inspected, and passed the qualification inspections specified in the general specification, unless otherwise specified herein, and has been listed on or approved for listing on the qualified products list.

3.2 Requirements - All requirements shall be in accordance with the general specification MIL-S-81018 except as otherwise specified herein.

3.3 Design and construction - The design and construction of the RSSK-1A rigid seat kit shall conform to that specified in the general specification MIL-S-81018, except as otherwise specified herein, and shall be in accordance with the drawings listed on the Naval Air Systems Command index drawing 67A73D1. The optional methods of manufacture covered in some of the drawings shall not compromise the performance or interchangeability requirements specified in the general specification and herein.

3.3.1 Operation - The operation of the RSSK-1A rigid seat survival kit shall be as specified in the general specification.

3.3.2 Components - The RSSK-1A rigid seat survival kit shall consist of the components specified in the general specification which are the following:

(a) Basic container
(b) Contoured comfort cushion
(c) Ejection seat to kit attaching hardware and webbing assembly.
(d) Kit to integrated torso harness attaching hardware and webbing assembly.

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(e) Operating handle and lid unlock assembly
(f) Survival equipment bag
(g) Container retaining lanyard assembly
(h) Life raft protective cover
(i) Emergency bailout oxygen system

3.3.2.1 Basic container - The basic container of the RSSK-1A rigid seat survival kit shall be designed to conform to the requirements specified in the general specification and shall be in accordance with the applicable drawings listed in the RSSK-1A index drawing.

3.3.2.2 Contoured comfort cushion - The contoured comfort cushion of the RSSK-1A rigid seat survival kit shall conform to the requirements specified in the general specification and shall be in accordance with the applicable drawings listed in the RSSK-1A index drawing.

3.3.2.3 Ejection seat to kit attaching hardware and webbing assembly - The ejection seat to kit attaching hardware of the RSSK-1A rigid seat survival kit shall be in accordance with the applicable drawings listed in the RSSK-1A index drawing. The strap assemblies shall retain the RSSK-1A seat kit in the applicable ejection seat during normal flight operations including negative "g" loads on the seat kit up to four "g's", without failure of the strap webbing or hardware.

3.3.2.4 Kit to integrated torso harness attaching hardware and webbing assembly - The kit to integrated torso harness attaching hardware and webbing assembly of the RSSK-1A rigid seat survival kit shall conform to the requirements specified in the general specification and shall be in accordance with the applicable drawing listed in the RSSK-1A index drawing.

3.3.2.5 Operating handle and lid unlock assembly - The operating handle and lid unlock assembly of the RSSK-1A rigid seat survival kit shall conform to the requirements specified in the general specification and shall be in accordance with the applicable drawing listed in the RSSK-1A index drawing.

3.3.2.6 Survival equipment bag - The survival equipment bag of the RSSK-1A rigid seat survival kit shall conform to the requirements specified in the general specification and shall be in accordance with the applicable drawing listed in the RSSK-1A index drawing.

3.3.2.7 Container retaining lanyard assembly - The container retaining lanyard assembly of the RSSK-1A rigid seat survival kit shall conform to the requirements specified in the general specification and shall be in accordance with the applicable drawing listed in the index RSSK-1A drawing.

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3.3.2.8 Life raft protective cover - The RSSK-1A rigid seat survival kit shall include a life raft protective cover which shall be in accordance with the applicable drawing listed in the RSSK-1A index drawing.

3.3.2.9 Emergency bailout oxygen system - Unless otherwise specified herein the emergency bailout oxygen system of the RSSK-1A rigid seat survival kit shall conform to the requirements specified in the general specification and shall be in accordance with the applicable drawings listed in the RSSK-1A index drawing.

3.3.2.9.1 Components - The RSSK-1A emergency oxygen system shall consist of the following components:

(a) Emergency oxygen cylinder assembly
(b) Oxygen manifold assembly
(c) Actuation mechanism
(d) Composite disconnect block assembly
(e) Electrical communication system

3.3.2.9.1.1 Emergency oxygen cylinder assembly - The emergency oxygen cylinder assembly of the RSSK-1A rigid seat survival kit shall conform to the requirements specified in the general specification and shall be in accordance with the applicable drawing listed in the RSSK-1A index drawing.

3.3.2.9.1.2 Oxygen manifold assembly - The oxygen manifold assembly of the RSSK-1A rigid seat survival shall be in accordance with the applicable drawings listed in the RSSK-1A index drawing. The oxygen manifold assembly shall incorporate the following components.

(a) Emergency type oxygen reducer, (on-off type)
(b) Filler check valve
(c) Cylinder pressure gage
(d) Relief valve
(e) Check valve

These components shall conform to the applicable requirements specified in the general specification.

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3.3.2.9.1.3 Actuation mechanism- The actuation mechanism of the RSSK-1A rigid seat survival kit shall conform to the requirements specified in the general specification and shall be in accordance with the applicable drawings listed in the RSSK-1A index drawing.

3.3.2.9.1.4 Composite disconnect block assembly - The composite disconnect block assembly of the RSSK-1A rigid seat survival kit shall be in accordance with the applicable drawings listed in the RSSK-1A index drawing. The composite disconnect block assembly shall consist of an upper block assembly, an intermediate block assemblt, a lower block assembly, and a manual release to actuate a quick disconnect of the block assemblies. The oxygen system and electrical communications shall be completed through the composite disconnect block assembly.

3.3.2.9.1.5 Electrical communications system - The electrical communications system of the RSSK-1A rigid seat survival kit shall be contained in the composite block disconnect assembly and shall be compatible and integrate with the aircraft-system through the communication leads of the applicable hoses.

3.3.3 Design load requirements - Unless otherwise specified herein, the design load requirements of the RSSK-1A rigid seat survival kit shall be as specified in the general specification.

3.3.3.1 Ejection seat to kit attaching hardware and webbing assembly - The ejection seat to kit attaching hardware and webbing assembly shall be capable of sustaining a collective ultimate tension load of 6500 pounds without failure of the attachment hardware or webbing, when tested as specified in Section 4. The four kit retaining strap assemblies shall each be capable of sustaining an ultimate tension load of 100 pounds without failure of the webbing or attaching hardware when tested as specified in Section 4.

3.3.3.2 Kit to integrated torso harness attaching hardware and webbing assembly - The kit to integrated torso harness attaching hardware and webbing assembly shall be capable of sustaining a collective ultimate tension load of 2500 pounds without failure of the attachment hardware or webbing, when tested as specified in Section 4.

3.4 Performance - The RSSK-1A rigid seat survival kit shall meet all the performance requirements of the general specification when tested as specified therein and as otherwise specified herein.

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3.5 Size - The RSSK-1A rigid seat survival kit shall conform to the dimensions specified in the applicable drawings listed in the RSSK-1A index drawing.

3.6 Weight - The weight of the RSSK-1A rigid seat survival kit including the emergency bailout oxygen system but not including the survival equipment shall be 28.1 pounds. The approximate weight of the complete seat kit including the survival equipment, shall be 48 pounds.

3.6.1 Center of gravity - The center of gravity of the RSSK-1A rigid seat survival kit including the emergency bailout oxygen system and the survival equipment package shall be specified by the Aerospace Crew Equipment Laboratory.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The examination and testing of the RSSK-1A rigid seat survival kit shall be in accordance with the examinations and tests specified in the general specification and shall be classified as follows:

(a) Qualification inspection - Qualification inspection consists of the examinations and tests performed on samples submitted for approval as qualified products.

(b) First article inspection - First article inspection consists of examinations and tests performed on the RSSK-1A rigid seat survival kit which are representative of the production item after award of a contract to determine that the production item meets the requirements of the general specification and those specified herein.

(c) Quality conformance inspection - Quality conformance inspection consists of examinations and tests performed on individual products or lots with the requirements set forth in the general specification and those specified herein.

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4.3 Qualification inspection - The qualification inspection of the RSSK-1A rigid seat survival kits shall consist of examinations and tests for all of the requirements of the general specification and as otherwise specified herein.

4.3.1 Qualification samples - The qualification samples shall be as specified in the general specification.

4.4 First article inspection - The first article inspection of the RSSK-1A rigid seat survival kits shall consist of examinations and tests for all of the requirements of the general specification and those herein as deemed necessary by the testing activity.

4.4.1 First article samples - Unless otherwise specified the first article samples shall be as specified in the general specification.

4.5 Quality conformance inspection - The quality conformance inspection for the RSSK-1A rigid seat survival kit shall consist of the same examinations and tests as specified in the general specification except as otherwise specified herein. The following tests shall be included as part of the quality conformance inspection:

Lower disconnect lanyard operation
Individual disconnect tests
Pressure drop

4.5.1 Sampling - The sampling of the RSSK-1A rigid seat survival kit shall be as specified in the general specification.

4.5.1.1 Sampling for tests and examinations of the RSSK-1A rigid survival kit - The following tests shall be made a part of those listed in Table I of the general specification:

INSPECTION	CLASSIFICATION OF CHARACTERISTIC	METHOD PARAGRAPH	SAMPLE SIZE	ACCEPTANCE CRITERIA
Lower disconnect lanyard operation	Critical	4.7.2	Inspection Level S-4 1/	Acceptance number zero rejection number 1
Individual disconnect tests	Critical	4.7.6.1	Inspection Level S-4 1/	Acceptance number zero rejection number 1
Pressure drop	Critical	4.7.4	Inspection Level S-4 1/	Acceptance number zero rejection number 1

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4.6 Test conditions - The requirements for test conditions shall be as specified in the general specification.

4.7 Inspection methods - The inspection methods for the RSSK-1A rigid seat survival kit shall be as specified in the general specification except as otherwise noted herein.

4.7.1 Kit actuation test - The kit actuation test shall be as specified in the general specification with the exception that sub-paragraph (a) is not applicable. The weight of the kit shall be 48 pounds.

4.7.2 Lower disconnect lanyard operation test - The lower disconnect assembly shall be disengaged from the container by a pull on the lanyard both with and without the upper block engaged with the container. The emergency oxygen release actuation shall occur only if the upper block is engaged with the container.

4.7.3 Load tests -

4.7.3.1 Bending test (Intermediate Block) - With the container closed and locked and placed as specified in the vertical compression test of the general specification, a load of 300 pounds shall be applied uniformally to the top of the intermediate block. The load shall be applied in a direction normal to the plane of the upper surface of the block for a period of 5 seconds. There shall be no failure of the block-to-container attachment and no flexing or bending of the container when checked by visual, non-instrument means.

4.7.3.2.1 Ejection seat to kit attaching hardware and webbing assembly static load test - The ejection seat to kit attaching hardware and webbing assembly (not to include the kit retaining straps) shall be tested as specified in the general specification, static load test.

4.7.3.2.1.1 Kit retaining strap assembly static load test - Each kit retaining strap assembly (4 straps) shall be subjected to a tensile load of 100 pounds without failure of the webbing or attachment hardware.

4.7.3.2.2 Kit to integrated torso harness attaching hardware and webbing assembly static load test - This test shall be conducted as specified in 4.7.3.2.1 with the exception that a suitable means of attaching the RSSK-1A kit shall be used to apply the load in the testing machine. The method of attachment shall be such that the loads imposed will be on the kit to integrated torso harness attaching hardware and webbing assembly. The load applied shall be 2500 pounds and there shall be no failure of the attaching hardware or webbing.

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4.7.4 Pressure drop through complete disconnect assembly with hoses - Inlet pressures of 40, 50, 70, and 90 psi are to be applied to the complete assembly. Flows are to be varied at each inlet pressure from zero to 140 l/m (STP) in 10 l/m increments. The pressure drop through the complete disconnect assembly including hoses shall not exceed 5 psi.

4.7.5 Electrical continuity - With the upper, intermediate, and lower disconnect blocks engaged, there shall be electrical continuity throughout the composite disconnect assembly.

4.7.6 Composite disconnect test - In addition to the tests performed on the composite disconnect blocks as part of the survival kit, the following tests shall be required on the lower, intermediate and upper disconnect block assemblies independent of the survival kit.

4.7.6.1 Individual disconnect tests - The composite disconnect shall be subjected to the following tests:

4.7.6.1.1 Disconnect leakage - The three block assemblies of the composite disconnect shall be engaged and pressurized by sealing the end of the upper block and the oxygen port on the side of the intermediate block, and applying pressure to the side of the lower block. All ports shall be pressurized simultaneously by applying 150 psi to the oxygen port, 40 psi to the anti-g port and 10 psi to the ventilation air port. The units shall be submerged in water and no evidence of leakage shall be observed.

4.7.6.1.2 Check valve gas leakage - The three block assemblies shall be separated and the check valves in each shall be tested as follows:

4.7.6.1.2.1 Upper and intermediate blocks - On the oxygen and ventilation air check valves, apply pressure in the direction opposite to normal gas flow. Valves shall be unsatisfactory if there is any leakage with 50 and 80 psi pressure on the oxygen valve and 0.5 and 3.6 psi pressure on the ventilation air valve.

4.7.6.1.2.2 Lower block - On the oxygen, ventilation air, and anit-g air check valves, apply pressure in the direction of normal gas flow. Valves shall be unsatisfactory if there is any leakage with 60 and 120 psi pressure on the oxygen valve, 0.5 and 3.6 psi on the ventilation air valve, and 15 psi on the anti-g air valve.

4.7.6.1.3 Check valve water leakage - All upper block check valves shall be tested for water leakage in the direction of normal gas flow at hydrostatic heads of 4, 8, 12, and 16 feet of water. The valve shall be unsatisfactory if any leakage is observed after one minute at each exposure. This test may be conducted with gas at equivalent pressures, in which case, leakage shall be zero as indicated on a flowmeter.

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4.7.6.1.4 Engaging and disengaging force - The composite disconnect shall be tested for engaging and disengaging forces. When the disconnect devices are not actuated, a pull of 250 pounds minimum shall not disengage the upper and/or lower block from the intermediate block. A force of 20 ± 5 pounds, applied as an engaging force shall lock the lower block into the intermediate block, or applied as a pull on the lanyard shall unlock it. When the lower block is locked in place, a force of 20 ± 5 pounds shall be sufficient to engage and lock the upper block into the intermediate block. A force of 15 ± 5 pounds shall be sufficient to release the locking device to permit disengagement of the upper block by the spring loaded ejector mechanism.

4.7.6.1.5 Separation sequence test - It shall be ascertained that the electrical pins engage after the oxygen check valve is engaged and that the electrical pins disengage prior to oxygen disengagement.

4.7.6.2 Endurance test - The assembled composite disconnect shall be sealed and pressurized as described in 4.7.6.1.1 at a rate of not less than 10 psi per second. The disconnect shall be capable of withstanding 1,000 pressurization cycles without separating. The composite disconnect shall then be subjected to an engaging and disengaging endurance test. With the Intermediate block maintained at 70° ± 2°F, 500 cycles shall be performed with the upper block at 70° ± 2°F and 500 cycles with the lower block at 70° ± 2°F. With the intermediate block maintained at 160° ± 2°F, 250 cycles shall be performed with the upper block at 160° ± 2°F and 250 cycles with the upper block at 70° ± 2°F. With the intermediate block maintained at -65° ± 2°F, 250 cycles shall be performed with the upper block at -65° ± 2°F and 250 cycles with the upper block at 70° ± 2°F. The disconnect shall then be subjected to all the tests of 4.7.6.1. and shall meet all the requirements thereof, except that the disengaging forces may be reduced no more than 10 per cent of the disengaging forces specified in 4.7.6.1.4.

4.7.7 Manual release cable positive lock indicator test - (Endurance) - The manual release cable shall be cycled 1,000 times at the rate of one cycle every three seconds. One cycle constitutes an engagement and disengagement of the upper block assembly. The lock indicator shall be evident at each cycle. After endurance cycling, the cable assembly shall be examined for distortion, fraying, and hang up. There shall be not evidence of any failures.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the requirements as specified in the general specification.

6. NOTES

6.1 All notes specified in the general specification are applicable to this specification.

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